FEA HW2

Ben Sarfati 885573816

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1

Given

$$\begin{cases} -\frac{d}{dx}\left(EA\frac{du}{dx}\right) = \rho A\omega^2 x \\ u(0) = 0 \\ EA\frac{du}{dx}\Big|_{x=L} = 0 \end{cases}$$

$$-\frac{d}{dx}\left(EA\frac{du}{dx}\right) = \rho A\omega^2 x$$

Integrating and applying boundary conditions twice over:

$$\Rightarrow -\left(EA\frac{du}{dx}\right) = \rho A\omega^2\frac{x^2}{2} + const.$$

$$EA\frac{du}{dx}\Big|_{x=L} = 0 \Rightarrow -\left(EA\frac{du}{dx}\right) = \rho A\omega^2\frac{x^2}{2} - \rho A\omega^2\frac{L^2}{2}$$

$$\Rightarrow u = \frac{\rho\omega^2}{2E}\left((-\frac{1}{3})x^3 + (L^2)x\right) + const.$$

$$u(0) = 0 \Rightarrow u = \frac{\rho\omega^2}{2E}\left((-\frac{1}{3})x^3 + (L^2)x\right)$$

$$\boxed{u(x) = \frac{\rho\omega^2 x}{2E}\left(L^2 - \frac{x^2}{3}\right)}$$

2

To apply the least squares method we first approximate the numerical solution as a sum of the Lagrange polynomial basis functions in the following manner:

$$\mathbf{x^T} = \begin{bmatrix}0 & \frac{L}{3} & L\end{bmatrix}$$

$$\mathcal{L}_1(x) = \frac{(x-\frac{L}{3})(x-L)}{\frac{L^2}{3}} = \frac{3}{L^2}(x-\frac{L}{3})(x-L)$$

$$\mathcal{L}_2(x) = \frac{x(x-L)}{-\frac{2L^2}{9}} = -\frac{9}{2L^2}x(x-L)$$

$$\mathcal{L}_3(x) = \frac{x\left(x-\frac{L}{3}\right)}{\frac{2L^2}{3}} = \frac{3}{2L^2}x(x-\frac{L}{3})$$

$$\tilde{u}(x) = u(0)\mathcal{L}_1(x) + u(\frac{L}{3})\mathcal{L}_2(x) + u(L)\mathcal{L}_3(x)$$

$$\tilde{u}(x) = 0 - \frac{9}{2L^2}x(x-L)u_2 + \frac{3}{2L^2}x(x-\frac{L}{3})u_3$$

$$\frac{d\tilde{u}(x)}{dx} = -\frac{9}{2L^2}(2x-L)u_2 + \frac{3}{2L^2}(2x-\frac{L}{3})u_3$$
$$\frac{d^2\tilde{u}(x)}{dx^2} = -\frac{9}{L^2}u_2 + \frac{3}{L^2}u_3 = \frac{-9u_2+3u_3}{L^2}$$

We can then write the residuals in the domain and on the Neumann boundary as

$$R = -\frac{d}{dx}(EA\frac{d\tilde{u}}{dx}) - \rho A\omega^2 x = -EA\frac{d^2\tilde{u}}{dx^2} - \rho A\omega^2 x = \frac{3EA}{L^2}(3u_2-u_3) - \rho A\omega^2 x$$

$$r = EA\frac{d\tilde{u}}{dx}\bigg|_{x=L} = EA(-\frac{9}{2L^2}(L)u_2 + \frac{3}{2L^2}(\frac{5L}{3})u_3) = \frac{EA}{2L}(5u_3-9u_2)$$

Now applying the least squares method to both residuals R and r:

$$\int_0^L R\frac{\partial R}{\partial u_2}dx + r\frac{\partial r}{\partial u_2} = \int_0^L (\frac{3EA}{L^2}(3u_2-u_3) - \rho A\omega^2 x)(\frac{9EA}{L^2})dx + (\frac{EA}{2L}(5u_3-9u_2))(-\frac{9EA}{2L}) = 0$$

$$\frac{27E^2A^2}{L^3}(3u_2-u_3) - \frac{9E^2A^2}{4L^2}(5u_3-9u_2) = \frac{9\rho\omega^2 EA^2}{2}$$

$$\int_0^L R\frac{\partial R}{\partial u_3}dx + r\frac{\partial r}{\partial u_3} = \int_0^L (\frac{3EA}{L^2}(3u_2-u_3) - \rho A\omega^2 x)(-\frac{3EA}{L^2})dx + (\frac{EA}{2L}(5u_3-9u_2))(\frac{5EA}{2L}) = 0$$

$$-\frac{9E^2A^2}{L^3}(3u_2-u_3) + \frac{5E^2A^2}{4L^2}(5u_3-9u_2) = -\frac{3\rho\omega^2 EA^2}{2}$$

Solving for the conveniently isolated linear combinations of the unknowns we get

$$\begin{cases} 5u_3 - 9u_2 = 0 \\ -u_3 + 3u_2 = \frac{\rho\omega^2 L^3}{6E} \end{cases}$$

It should be noted that by setting the boundary residual to zero we immediately would have arrived at the first equation. After solving we get

$$\boxed{\begin{cases} u_3 = \frac{\rho\omega^2 L^3}{4E} \\ u_2 = \frac{5\rho\omega^2 L^3}{36E} \\ u_1 = 0 \end{cases}}$$

The first nodal value of course results from the homogeneous dirichlet BC at 0.

3

Before applying the Galerkin method we define the residuals in the domain and on the Neumann boundary, introducing an approximate form of the solution $\tilde{u} = \phi_j a_j$:

$$R = -\frac{d}{dx}(EA\frac{d\tilde{u}}{dx}) - \rho A\omega^2 x = -\frac{d}{dx}(EA\frac{d\phi_j}{dx})a_j - \rho A\omega^2 x$$

$$r = EA\frac{d\tilde{u}}{dx}\bigg|_{x=L} = EA\frac{d\phi_j}{dx}a_j\bigg|_{x=L}$$

Now we demand

$$\int_0^L \phi_i R dx + \phi_i(L) r = 0$$

$$\Rightarrow \int_0^L \phi_i\Big[-\frac{d}{dx}(EA\frac{d\phi_j}{dx})a_j - \rho A\omega^2 x\Big]dx + \phi_i(L)\Big[EA\frac{d\phi_j}{dx}a_j\Big]_{x=L} = 0$$

$$\Rightarrow \int_0^L \phi_i\Big[-\frac{d}{dx}(EA\frac{d\phi_j}{dx})a_j\Big]dx + \phi_i(L)\Big[EA\frac{d\phi_j}{dx}a_j\Big]_{x=L} = \int_0^L \phi_i\Big[\rho A\omega^2 x\Big]dx$$

$$\phi_i\Big[-\frac{d}{dx}(EA\frac{d\phi_j}{dx})a_j\Big] = \frac{d}{dx}\Big[-\phi_i(EA\frac{d\phi_j}{dx})a_j\Big] + \frac{d\phi_i}{dx}(EA\frac{d\phi_j}{dx})a_j$$

$$\Rightarrow \int_0^L \frac{d\phi_i}{dx}(EA\frac{d\phi_j}{dx})a_j dx + \Big[-\phi_i(EA\frac{d\phi_j}{dx})a_j\Big]_0^L + \phi_i(L)\Big[EA\frac{d\phi_j}{dx}a_j\Big]_{x=L} = \int_0^L \phi_i\Big[\rho A\omega^2 x\Big]dx$$

We can see that the right boundary terms cancel out which is expected considering a homogeneous Neumann BC. We also demand $i \neq 1 \Rightarrow \phi_i(0) = 0$ in order to get rid of the left boundary term $\forall i > 1$. We are left with

$$\boxed{i \neq 1 \Rightarrow \int_0^L \frac{d\phi_i}{dx}(EA\frac{d\phi_j}{dx})a_j dx = \int_0^L \phi_i\Big[\rho A\omega^2 x\Big]dx}$$

This is the weak form of the problem.

4

We would now like to generalize the problem to a master element. We introduce N nodal points x_j between which lie $n = N - 1$ regions called elements, and the following transformation $\forall e \leq n \in \mathbb{N}$ (e is not in Einstein notation):

$$\xi^e(x) = \frac{2}{h^e}x - \frac{x_e + x_{e+1}}{h^e};\ x(\xi^e) = \frac{h^e}{2}\xi^e + \frac{x_e + x_{e+1}}{2}$$

Where

$$h^e = x_{e+1} - x_e$$
$$J^e = \frac{dx}{d\xi^e} = \frac{h^e}{2}$$

For each element there are only two shape functions so we will define

$$\begin{cases} \phi_1^e(x) = \phi_e(x) \\ \\ \phi_2^e(x) = \phi_{e+1}(x) \end{cases}$$

$$\begin{cases} \hat{\phi}_1(\xi^e) = \frac{1}{2}(1 - \xi^e) \\ \\ \hat{\phi}_2(\xi^e) = \frac{1}{2}(1 + \xi^e) \end{cases}$$

$$\begin{cases} B_1(\xi^e) = \frac{d\hat{\phi}_1(\xi^e)}{d\xi^e} = -\frac{1}{2} \\ \\ B_2(\xi^e) = \frac{d\hat{\phi}_2(\xi^e)}{d\xi^e} = \frac{1}{2} \end{cases}$$

Note that

$$\phi_1^e(x(\xi^e)) = \hat{\phi}_1(\xi^e)$$
$$\phi_2^e(x(\xi^e)) = \hat{\phi}_2(\xi^e)$$

Therefore

$$\frac{d\phi_i^e(x)}{dx}\Big|_{x(\xi^e)} = \frac{d\phi_i^e(x(\xi^e))}{d\xi^e}\frac{d\xi^e}{dx} = \frac{d\hat{\phi}_i(\xi^e)}{d\xi^e}\frac{1}{J^e} = B_i(\xi^e)\frac{2}{h^e}$$

The integral over the domain can be split into a sum of integrals over each element, and since the only nonzero shape functions over an element are its own, we can say

$$K_{ij}^g = \int_0^L \frac{d\phi_i}{dx}(EA\frac{d\phi_j}{dx})dx$$
$$K_{ij}^e = \int_{x_e}^{x_{e+1}} \frac{d\phi_i^e}{dx}(EA\frac{d\phi_j^e}{dx})dx$$
$$\mathbf{K^g} = \mathop{A}_{e=1}^{n} \mathbf{K^e}$$

Where A is the special matrix summation specific in tutorial 3 of the course. To finish writing the integral form of the stiffness matrix for a master element we simplify to

$$\boxed{K_{ij}^e = \int_{-1}^1 \frac{1}{J^e}B_i EA \frac{1}{J^e}B_j J^e d\xi^e = \frac{2}{h^e}\int_{-1}^1 B_i EA B_j d\xi^e}$$

5

We would now like to calculate the above matrix explicitly. Since our integrand is a **constant** function we require $2p-1 \geq 0 \Rightarrow \mathbb{N} \ni p \geq 1$ Gauss points for **exact** integral value using Gaussian quadrature (we could also simply instruct a solver to multiply the integrand by the length of the master element).

$$\xi_k^e = 0; \; w_k = 2$$

$$K_{ij}^e = \sum_{k=1}^{1} w_k \frac{2}{h^e} B_i E A B_j$$

$$\Rightarrow \boxed{\mathbf{K^e} = \frac{EA}{h^e}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}}$$

6

To write the integral form of the forcing vector we introduce a Lagrange polynomial interpolation which exactly represents the linear forcing forcing function:

$$\begin{cases} x_1^{\mathcal{L}} = 0 \\ x_2^{\mathcal{L}} = L \end{cases} ; \begin{cases} \mathcal{L}_1 = \frac{x-L}{-L} \\ \mathcal{L}_2 = \frac{x}{L} \end{cases} ; \; f(x) = \rho A \omega^2 x; \; f(x) = f(x_j^{\mathcal{L}})\mathcal{L}_j(x)$$

$$\int_0^L \phi_i \left[\rho A \omega^2 x\right] dx = \left[\int_0^L \phi_i \mathcal{L}_j(x) dx\right] f(x_j^{\mathcal{L}})$$

This process is a bit redundant, and defeats the purpose of the master element transformation. Alternatively, we could interpolate the the forcing function with local Lagrange polynomials, which ends up being identical to approximating the forcing function using the already established linear basis functions. In this way:

$$\int_0^L \phi_i \left[\rho A \omega^2 x\right] dx \approx \left[\int_0^L \phi_i \phi_j dx\right] \rho A \omega^2 x_j$$

To calculate the "mass matrix" we apply the same process with the same reasoning as we do to the stiffness matrix from before:

$$M_{ij}^g = \int_0^L \phi_i \mathcal{L}_j(x) dx$$

$$M_{ij}^e = \int_{x_e}^{x_{e+1}} \phi_i^e \mathcal{L}_j(x) dx$$

$$\mathbf{M^g} = \mathop{A}_{e=1}^{n} \mathbf{M^e}$$

$$M_{ij}^e = \int_{-1}^{1} \hat{\phi}_i(\xi^e)\mathcal{L}_j(x(\xi^e)) J^e d\xi^e = \frac{h^e}{2}\int_{-1}^{1} \hat{\phi}_i(\xi^e)\mathcal{L}_j(x(\xi^e)) d\xi^e$$

(Side note: these integrals can be done analytically according to the assignment but were already done numerically in this manner, despite this being far more difficult. The result is still exact.)
For Gaussian quadrature we need $2p-1 \geq 2 \Rightarrow \mathbb{N} \ni p \geq 2$ Gauss points **for exact integral value** since our integrand is quadratic:

$$\xi_1^e = \frac{1}{\sqrt{3}}; \; \xi_2^e = -\frac{1}{\sqrt{3}}; \; w_k = 1$$

$$M_{ij}^e = \sum_{k=1}^{2} w_k \hat{\phi}_i(\xi_k^e)\mathcal{L}_j(x(\xi_k^e))$$

$$M_{11}^e = \sum_{k=1}^{2} w_k \hat{\phi}_1(\xi_k^e)\mathcal{L}_1(x(\xi_k^e)) = \frac{1}{2}(1-\frac{1}{\sqrt{3}})(\frac{\frac{h^e}{2}\frac{1}{\sqrt{3}} + \frac{x_e+x_{e+1}}{2} - L}{-L}) + \frac{1}{2}(1+\frac{1}{\sqrt{3}})(\frac{-\frac{h^e}{2}\frac{1}{\sqrt{3}} + \frac{x_e+x_{e+1}}{2} - L}{-L})$$

$$= \frac{\frac{x_e+x_{e+1}}{2} - L}{-L} - \frac{1}{\sqrt{3}}\frac{\frac{h^e}{2}\frac{1}{\sqrt{3}}}{-L} = -\frac{1}{L}(\frac{x_e+x_{e+1}}{2} - L - \frac{h^e}{6})$$

$$M_{12}^e = \sum_{k=1}^{2} w_k \hat{\phi}_1(\xi_k^e)\mathcal{L}_2(x(\xi_k^e)) = \frac{1}{2}(1-\frac{1}{\sqrt{3}})(\frac{\frac{h^e}{2}\frac{1}{\sqrt{3}} + \frac{x_e+x_{e+1}}{2}}{L}) + \frac{1}{2}(1+\frac{1}{\sqrt{3}})(\frac{-\frac{h^e}{2}\frac{1}{\sqrt{3}} + \frac{x_e+x_{e+1}}{2}}{L})$$

$$= \frac{\frac{x_e+x_{e+1}}{2}}{L} - \frac{1}{\sqrt{3}}\frac{\frac{h^e}{2}\frac{1}{\sqrt{3}}}{L} = \frac{1}{L}(\frac{x_e+x_{e+1}}{2} - \frac{h^e}{6})$$

$$M_{21}^e = \sum_{k=1}^{2} w_k \hat{\phi}_2(\xi_k^e)\mathcal{L}_1(x(\xi_k^e)) = \frac{1}{2}(1+\frac{1}{\sqrt{3}})(\frac{\frac{h^e}{2}\frac{1}{\sqrt{3}} + \frac{x_e+x_{e+1}}{2} - L}{-L}) + \frac{1}{2}(1-\frac{1}{\sqrt{3}})(\frac{-\frac{h^e}{2}\frac{1}{\sqrt{3}} + \frac{x_e+x_{e+1}}{2} - L}{-L})$$

$$= \frac{\frac{x_e+x_{e+1}}{2} - L}{-L} + \frac{1}{\sqrt{3}}\frac{\frac{h^e}{2}\frac{1}{\sqrt{3}}}{-L} = -\frac{1}{L}(\frac{x_e+x_{e+1}}{2} - L + \frac{h^e}{6})$$

$$M_{22}^e = \sum_{k=1}^{2} w_k \hat{\phi}_2(\xi_k^e)\mathcal{L}_2(x(\xi_k^e)) = \frac{1}{2}(1+\frac{1}{\sqrt{3}})(\frac{\frac{h^e}{2}\frac{1}{\sqrt{3}} + \frac{x_e+x_{e+1}}{2}}{L}) + \frac{1}{2}(1-\frac{1}{\sqrt{3}})(\frac{-\frac{h^e}{2}\frac{1}{\sqrt{3}} + \frac{x_e+x_{e+1}}{2}}{L})$$

$$= \frac{\frac{x_e+x_{e+1}}{2}}{L} + \frac{1}{\sqrt{3}}\frac{\frac{h^e}{2}\frac{1}{\sqrt{3}}}{L} = \frac{1}{L}(\frac{x_e+x_{e+1}}{2} + \frac{h^e}{6})$$

$$\Rightarrow \mathbf{M^e} = \frac{1}{L}\begin{bmatrix} -(\frac{x_e+x_{e+1}}{2} - L - \frac{h^e}{6}) & (\frac{x_e+x_{e+1}}{2} - \frac{h^e}{6}) \\ -(\frac{x_e+x_{e+1}}{2} - L + \frac{h^e}{6}) & (\frac{x_e+x_{e+1}}{2} + \frac{h^e}{6}) \end{bmatrix}$$

The global forcing function is:

$$\mathbf{f} = \mathbf{f^e} = \begin{bmatrix} 0 \\ \rho A\omega^2 L \end{bmatrix}$$

It is the same as the local forcing function in this case because the Lagrange polynomials are global. The local forcing vector is then

$$\mathbf{F^e} = \mathbf{M^e f^e} = \rho A\omega^2 \begin{bmatrix} (\frac{x_e+x_{e+1}}{2} - \frac{h^e}{6}) \\ (\frac{x_e+x_{e+1}}{2} + \frac{h^e}{6}) \end{bmatrix}$$

Alternatively, with the alternative RHS approximation:

$$M_{ij}^g = \int_0^L \phi_i\phi_j dx$$

$$M_{ij}^e = \int_{x_e}^{x_{e+1}} \phi_i^e\phi_j^e dx$$

$$\mathbf{M^g} = \mathop{A}_{e=1}^{n} \mathbf{M^e}$$

$$M_{ij}^e = \int_{-1}^{1} \hat{\phi}_i(\xi^e)\hat{\phi}_j(\xi^e)J^e d\xi^e = \frac{h^e}{2}\int_{-1}^{1} \hat{\phi}_i(\xi^e)\hat{\phi}_j(\xi^e)d\xi^e$$

Once again for Gaussian quadrature we need $2p-1 \geq 2 \Rightarrow \mathbb{N} \ni p \geq 2$ Gauss points since our integrand is quadratic:

$$\xi_1^e = \frac{1}{\sqrt{3}};\ \xi_2^e = -\frac{1}{\sqrt{3}};\ w_k = 1$$

$$\frac{2}{h^e}M_{ij}^e = \sum_{k=1}^{2} w_k\hat{\phi}_i(\xi_k^e)\hat{\phi}_i(\xi_k^e)$$

$$\frac{2}{h^e}M_{11}^e = \sum_{k=1}^{2} w_k\hat{\phi}_1(\xi_k^e)\hat{\phi}_1(\xi_k^e) = \frac{1}{4}(1-\frac{1}{\sqrt{3}})^2 + \frac{1}{4}(1+\frac{1}{\sqrt{3}})^2 = \frac{1}{4}(1-2\frac{1}{\sqrt{3}}+\frac{1}{3}) + \frac{1}{4}(1+2\frac{1}{\sqrt{3}}+\frac{1}{3}) = \frac{2}{3}$$

$$\frac{2}{h^e}M_{12}^e = \sum_{k=1}^{2} w_k\hat{\phi}_1(\xi_k^e)\hat{\phi}_2(\xi_k^e) = \frac{1}{4}(1-\frac{1}{\sqrt{3}})(1+\frac{1}{\sqrt{3}}) + \frac{1}{4}(1+\frac{1}{\sqrt{3}})(1-\frac{1}{\sqrt{3}}) = \frac{1}{3}$$

$$\frac{2}{h^e}M_{21}^e = \sum_{k=1}^{2} w_k\hat{\phi}_1(\xi_k^e)\hat{\phi}_2(\xi_k^e) = \frac{1}{4}(1+\frac{1}{\sqrt{3}})(1-\frac{1}{\sqrt{3}}) + \frac{1}{4}(1-\frac{1}{\sqrt{3}})(1+\frac{1}{\sqrt{3}}) = \frac{1}{3}$$

$$\frac{2}{h^e}M_{22}^e = \sum_{k=1}^{2} w_k\hat{\phi}_1(\xi_k^e)\hat{\phi}_1(\xi_k^e) = \frac{1}{4}(1+\frac{1}{\sqrt{3}})^2 + \frac{1}{4}(1-\frac{1}{\sqrt{3}})^2 = \frac{1}{4}(1+2\frac{1}{\sqrt{3}}+\frac{1}{3}) + \frac{1}{4}(1-2\frac{1}{\sqrt{3}}+\frac{1}{3}) = \frac{2}{3}$$

$$\Rightarrow \mathbf{M^e} = \frac{h^e}{6}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

The global forcing function is simply $f_j = \rho A\omega^2 x_j$ so the local forcing function is

$$\mathbf{f^e} = \rho A\omega^2 \begin{bmatrix} x_e \\ x_{e+1} \end{bmatrix}$$

So the local forcing vector is

$$\mathbf{F^e} = \mathbf{M^e f^e} = \rho A\omega^2 \frac{h^e}{6}\begin{bmatrix} 2x_e + x_{e+1} \\ x_e + 2x_{e+1} \end{bmatrix}$$

7

Solving with $n = 2$:

$$\Rightarrow h^e = \frac{L}{n} = \frac{L}{2};\ x_j = (j-1)h^e = (j-1)\frac{L}{2}$$

$$\mathbf{K^g a} = \mathbf{F^g}$$

$$\mathbf{K^e} = \frac{EA}{h^e}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\Rightarrow \mathbf{K^g} = \frac{EA}{h^e}\begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\mathbf{F^e} = \rho A\omega^2 \frac{h^e}{6}\begin{bmatrix} 2x_e + x_{e+1} \\ x_e + 2x_{e+1} \end{bmatrix}$$

$$\Rightarrow \mathbf{F^g} = \rho A\omega^2 \frac{h^e}{6}\begin{bmatrix} 2x_1 + x_2 \\ x_1 + 4x_2 + x_3 \\ x_2 + 2x_3 \end{bmatrix}$$

Note that the system is incorrect as long as the dirichlet BC is not present:

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \frac{\rho(h^e\omega)^2}{6E}\begin{bmatrix} 0 \\ x_1 + 4x_2 + x_3 \\ x_2 + 2x_3 \end{bmatrix}$$

Moving the first column to the RHS:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \frac{\rho(h^e\omega)^2}{6E}\begin{bmatrix} 0 \\ x_1 + 4x_2 + x_3 \\ x_2 + 2x_3 \end{bmatrix} - 0\begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$

Simplifying:

$$\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} a_2 \\ a_3 \end{bmatrix} = \frac{\rho(h^e\omega)^2}{6E}\begin{bmatrix} x_1 + 4x_2 + x_3 \\ x_2 + 2x_3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} a_2 \\ a_3 \end{bmatrix} = \frac{\rho(h^e\omega)^2}{6E}\begin{bmatrix} x_1 + 5x_2 + 3x_3 \\ x_2 + 2x_3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} a_2 \\ a_3 \end{bmatrix} = \frac{\rho(h^e\omega)^2}{6E}\begin{bmatrix} x_1 + 5x_2 + 3x_3 \\ x_1 + 6x_2 + 5x_3 \end{bmatrix} = \frac{\rho(h^e\omega)^2 L}{6E}\begin{bmatrix} \frac{11}{2} \\ 8 \end{bmatrix}$$

$$\begin{cases} a_1 = 0 \\ a_2 = \frac{11\rho(h^e\omega)^2}{12E} \\ a_3 = \frac{4\rho(h^e\omega)^2}{3E} \end{cases}$$

Finally, our choice of shape functions allows us to make the simplification

$$\tilde{u} = \phi_j a_j \Rightarrow \tilde{u} = \begin{cases} \phi_j^e a_{e+j-1} & x_e \leq x \leq x_{e+1} \end{cases}$$

Again, this applies $\forall e$; e is not in Einstein notation.

$$\phi_1^1(x) = \hat{\phi}_1(\xi^1(x)) = \frac{1}{2}\left(1 - (\frac{2}{h^e}x - \frac{x_1 + x_2}{h^e})\right) = 1 - x$$

$$\phi_2^1(x) = \hat{\phi}_2(\xi^1(x)) = \frac{1}{2}\left(1 + (\frac{2}{h^e}x - \frac{x_1 + x_2}{h^e})\right) = x$$

$$\phi_1^2(x) = \hat{\phi}_1(\xi^2(x)) = \frac{1}{2}\left(1 - (\frac{2}{h^e}x - \frac{x_2 + x_3}{h^e})\right) = 2 - x$$

$$\phi_2^2(x) = \hat{\phi}_2(\xi^2(x)) = \frac{1}{2}\left(1 + (\frac{2}{h^e}x - \frac{x_2 + x_3}{h^e})\right) = x - 1$$

$$\Rightarrow \boxed{\tilde{u} = \begin{cases} \frac{11\rho(h^e\omega)^2}{12E}x & 0 \leq x \leq \frac{L}{2} \\ \frac{11\rho(h^e\omega)^2}{12E}(2 - x) + \frac{4\rho(h^e\omega)^2}{3E}(x - 1) & \frac{L}{2} \leq x \leq L \end{cases}}$$